PROJECT DOCUMENTATION

Project Title

Travel Guide AI – An Intelligent Travel Recommendation System

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Team Details

Team ID: LTVIP2026TMIDS88672

Team Size : 5

Team Leader : Kranthi Polanki

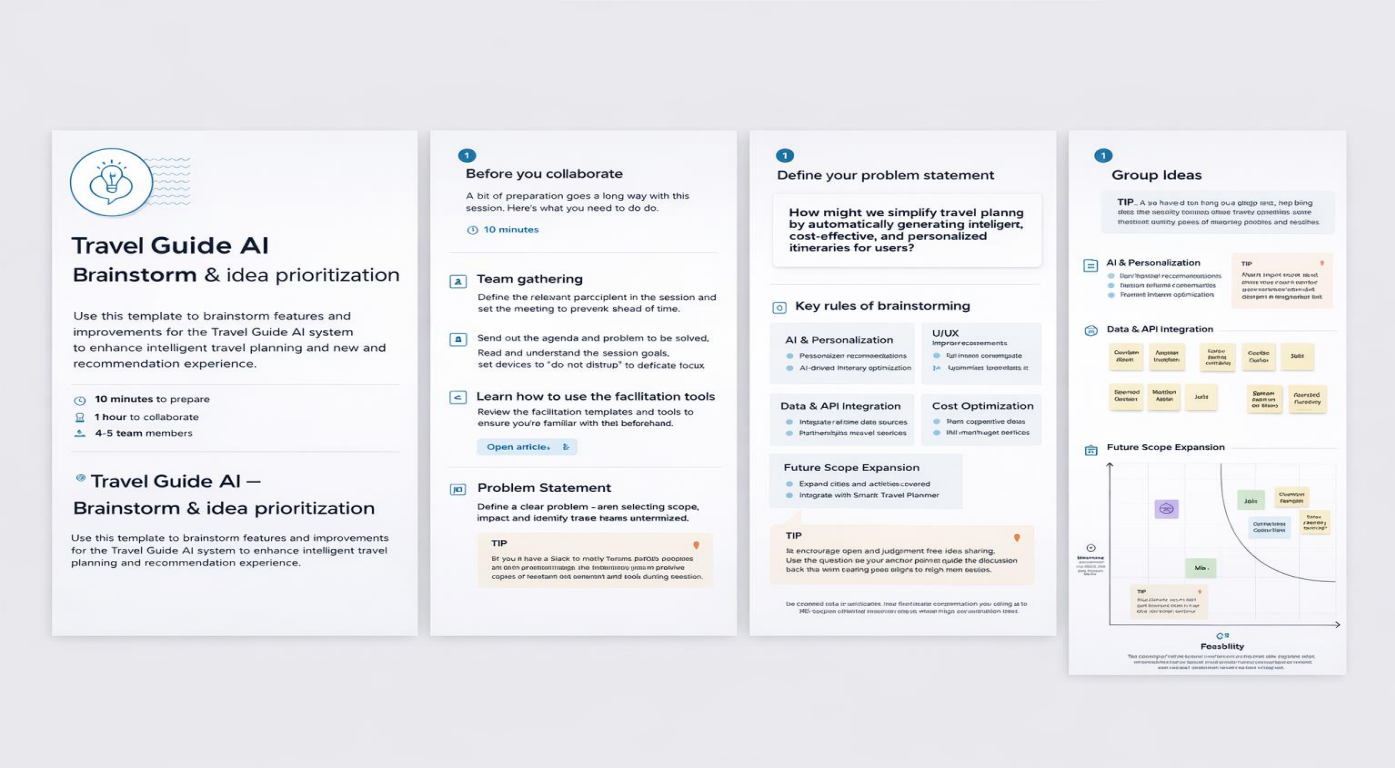
Team member : Bodanapu Venkata Keerthana

Team member : Meghashya Yarasani

Team member : Harshitha Raghava

Team member : Sulaiman Shaik

Phase 1: Brainstorming & Ideation



1.1 Objective

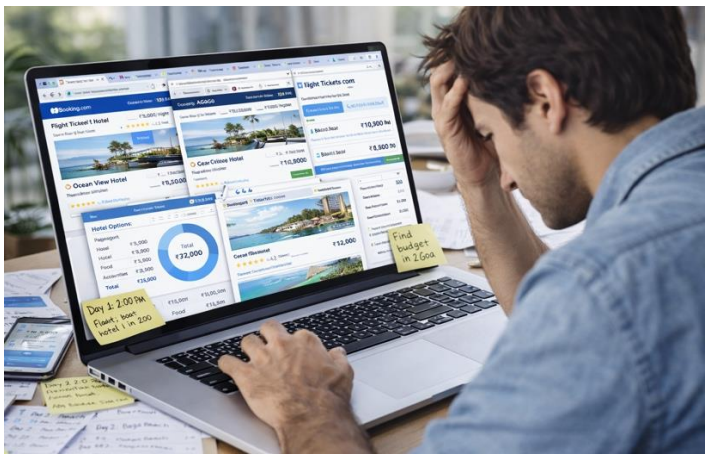
To design and develop an intelligent travel planning system that generates complete state-wise travel plans including destinations, cost estimation, recommended stay, and day-wise itinerary.

1.2 Problem Statement

Travelers often face difficulty selecting destinations, planning itineraries, estimating budgets, and gathering reliable information from multiple sources.

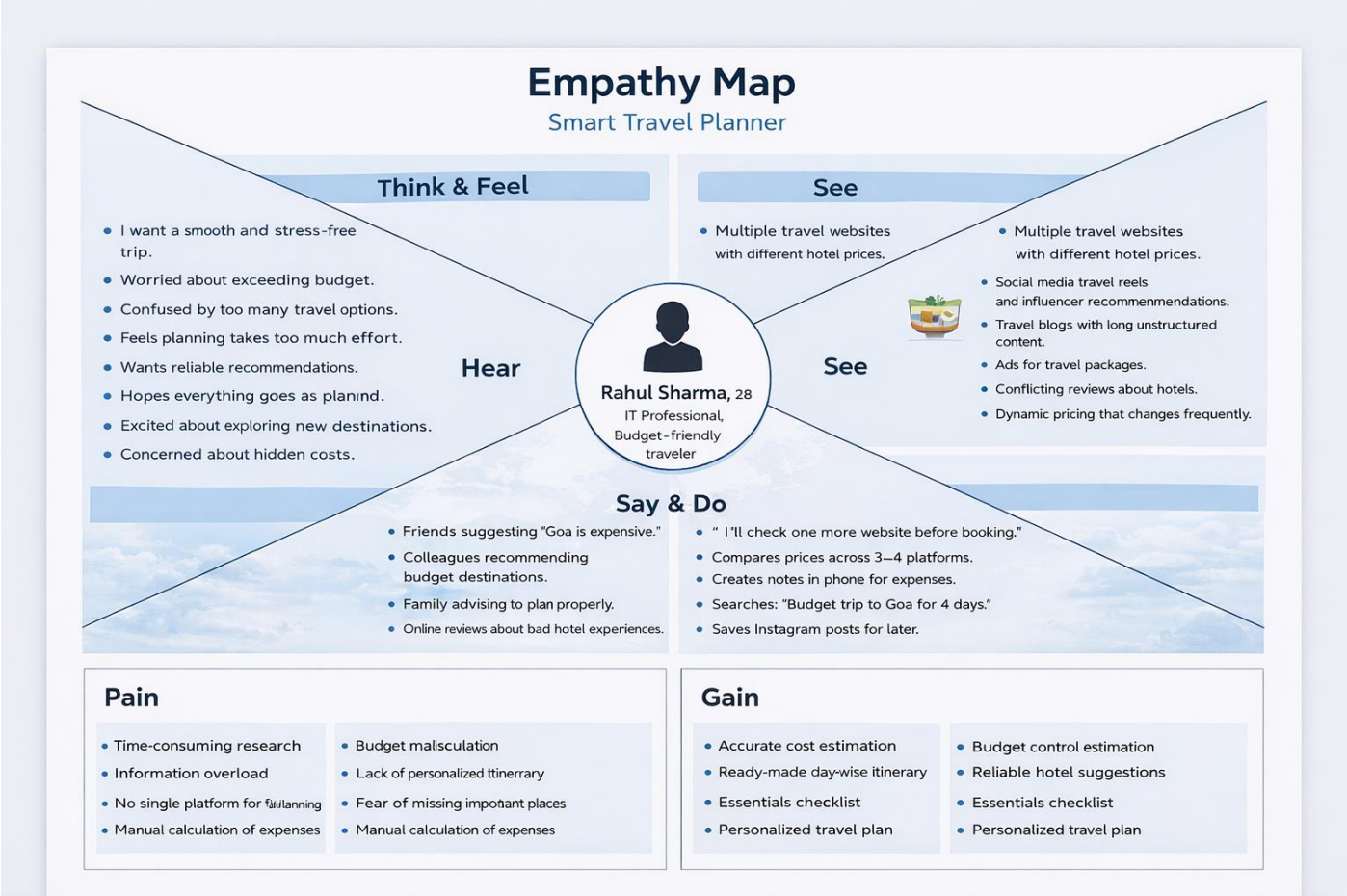
Customer Problem Statement:

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	**I am** a traveler planning a vacation within India.	**I'm trying to** choose a state, finalize destinations, estimate my travel budget, and create a proper day-wise itinerary.	**But** it takes too much time to search for destinations, compare hotel prices, calculate total expenses, and organize everything manually.	**Because** information is scattered across multiple websites, pricing is inconsistent, and there is no single platform that generates a complete state-wise travel plan automatically.	**Which makes me feel** confused, overwhelmed, and frustrated before even starting my trip.
PS-2	**I am** a working professional with limited time for trip planning.	**I'm trying to** quickly generate a well-structured travel plan including cost estimation, recommended stay, and essential items for my trip.	**But** I struggle to estimate total expenses accurately and often miss important travel details.	**Because** I lack reliable consolidated data, real-time cost insights, and personalized itinerary recommendations.	**Which makes me feel** stressed and uncertain about whether my trip will stay within budget.



1.3 Empathize & Discover

Empathy Map:



The Empathy Map was created to understand the user's travel planning challenges, emotions, and expectations. It helped identify key problems faced by travelers and ensured that the Smart Travel Planner is designed as a user-centered solution.

Key Pain Points:

- Time-consuming research
- Scattered travel information
- Budget miscalculation
- Lack of personalized itinerary
- Manual expense calculation
- Stress before booking

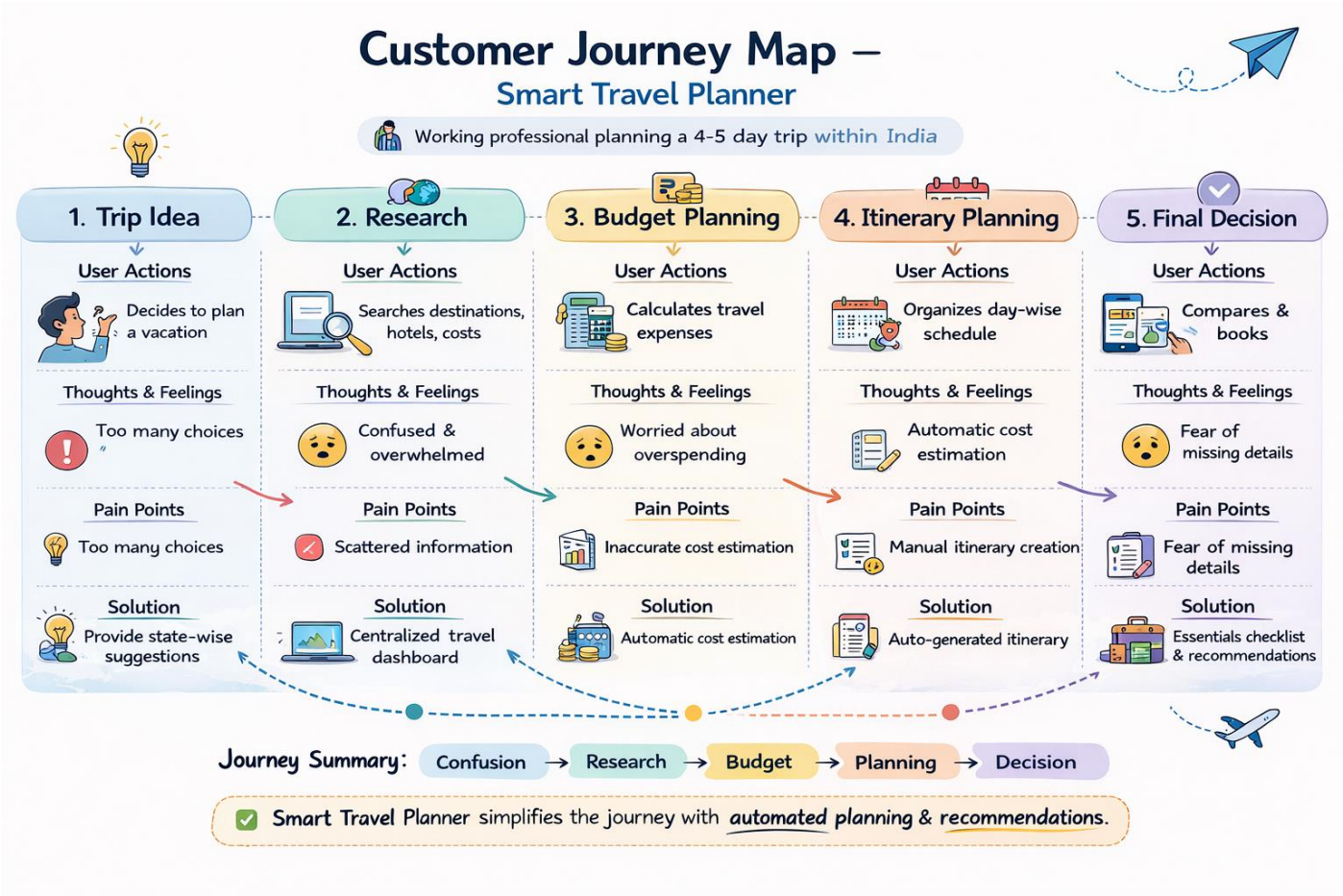
Key Gains:

- Accurate cost estimation
- Ready-made day-wise itinerary
- Reliable hotel recommendations
- Essentials checklist
- Time-saving planning
- Budget control

Key Insights from Ideation:

• Users struggle with scattered travel information
• Budget estimation is difficult
• Travelers need automated itinerary planning
• A centralized platform is required.

1.4 Customer Journey Map



Journey Summary:

The customer journey analysis shows that travelers move from excitement to confusion during research and budgeting stages. The Smart Travel Planner simplifies this journey by providing automated cost estimation, personalized itinerary generation, and centralized travel planning in one platform.

Key Insights from Ideation:

• Users struggle with scattered travel information
• Budget estimation is difficult without structured breakdown
• Travelers need automated itinerary planning
• A centralized intelligent travel platform is required

1.5 Proposed Solution

Develop a Smart Travel Planner that allows users to select a state, destination, travel type, and number of days to automatically generate cost estimation, hotel recommendations, essentials list, and day-wise itinerary.

Phase 2: Requirement Analysis

2.1 Functional Requirements

FR No	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Travel Input Selection	User selects State
		User selects Destination
		User selects Travel Type (Friends / Family / Solo / Couple)
		User selects Number of Days
FR-2	Input Validation	System validates all user inputs
FR-3	Travel Recommendation Engine	Generate recommended destinations based on state
		Suggest travel style based on travel type
FR-4	Cost Estimation	Calculate estimated budget range
		Provide cost breakdown (travel, stay, food, activities)
FR-5	Hotel Recommendation	Display Budget / Mid-range / Luxury options
FR-6	Itinerary Generation	Generate day-wise travel plan
		Include attractions and activity suggestions
FR-7	Essentials Checklist	Display travel essentials based on destination
FR-8	Output Display	Show structured travel plan in user-friendly format

2.2 Non-Functional Requirements

NFR No	Non-Functional Requirement	Description
NFR-1	Usability	The system must provide a simple and user-friendly interface.
NFR-2	Performance	Travel plan should be generated within a few seconds.
NFR-3	Reliability	The system must generate consistent and accurate recommendations.
NFR-4	Scalability	The system should support adding more states and destinations in future.
NFR-5	Maintainability	Code should be modular and easy to update.
NFR-6	Availability	The system should be accessible whenever the user accesses it.
NFR-7	Compatibility	The system should run smoothly on web browsers via Streamlit.

2.3 Technology Stack

Component	Technology Used	Purpose / Description
Programming Language	Python	Used to implement backend logic, cost calculation, and itinerary generation
Framework	Streamlit	Used to build interactive web-based user interface
Recommendation Logic	Rule-Based Recommendation System	Generates travel suggestions based on user inputs
Version Control	GitHub	Used for code management and collaboration
Deployment Environment	Localhost (Cloud-ready)	Application runs locally and can be deployed to cloud platforms in future

2.4 User stories

User Type	Epic	User Story No	User Story	Acceptance Criteria	Priority	Release
Customer	Travel Planning	US-1	As a user, I can select state, destination, travel type, and number of days to plan my trip.	System accepts inputs and proceeds to generate plan.	High	Sprint 1
Customer	Recommendation	US-2	As a user, I receive recommended attractions and hotels based on my selection.	System displays relevant suggestions correctly.	High	Sprint 1
Customer	Cost Estimation	US-3	As a user, I can view estimated total trip cost.	System shows clear cost breakdown.	High	Sprint 1
Customer	Itinerary Generation	US-4	As a user, I receive a structured day-wise travel itinerary.	System displays organized daily schedule.	High	Sprint 1

Phase 3: Project Design

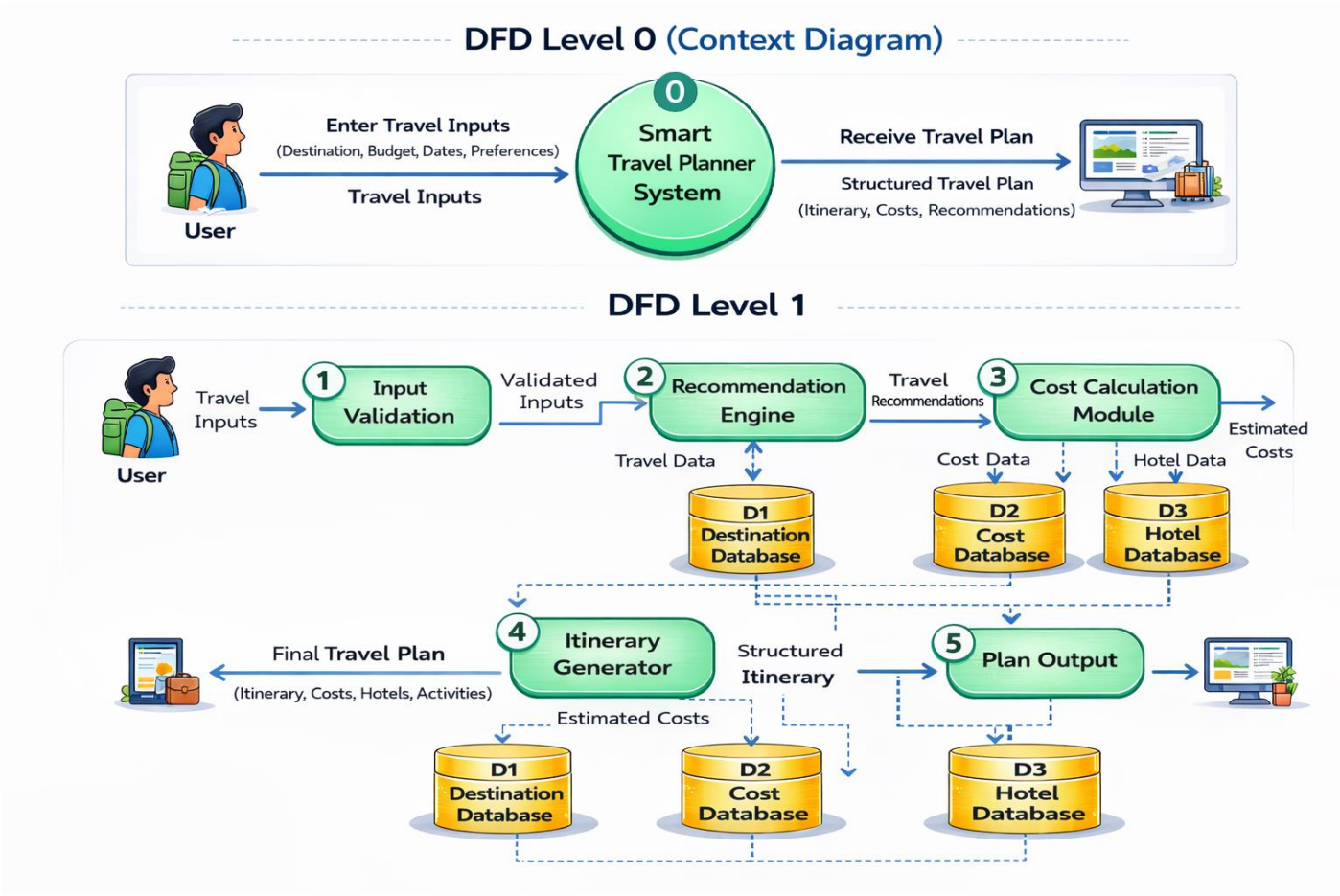
3.1 System Architecture

The Smart Travel Planner uses a modular architecture to process user inputs and generate a complete travel plan. The system includes Input Validation, Recommendation Engine, Cost Calculation Module, and Itinerary Generator. It uses destination, hotel, and cost datasets to provide accurate recommendations, budget estimation, and a structured day-wise itinerary.

User Input → Input Validation → Recommendation Engine → Cost Calculation → Itinerary Generation → Display Results

3.2 Data Flow Diagram (DFD)

The following Data Flow Diagram illustrates the overall data movement and internal processing of the Smart Travel Planner system.



Level 0 (Context Diagram) presents the overall interaction: the user enters travel details (destination, budget, dates, preferences), and the system generates a structured travel plan.

Level 1 explains the internal processes, including Input Validation, Recommendation Engine, Cost Calculation, and Itinerary Generation. The system uses Destination, Cost, and Hotel databases to produce recommendations, estimated costs, and a final travel itinerary.

This diagram clearly illustrates the flow of data from user input to final travel plan output.

Phase 4: Project Planning (Agile)

4.1 Product Backlog, Sprint Schedule and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Travel Input Selection	US-1	As a user, I can select state, destination, travel type, and number of days to plan my trip.	3	High	All
Sprint-1	Input Validation	US-2	As a system, I validate user inputs before generating the travel plan.	2	High	Backend Team
Sprint-2	Recommendation Engine	US-3	As a user, I receive recommended attractions based on my selected destination.	4	High	Backend Team
Sprint-2	Cost Estimation	US-4	As a user, I can view estimated total trip cost with proper cost breakdown.	4	High	Backend Team
Sprint-3	UI Development	US-5	As a user, I can interact with dropdowns, sliders, and generate the travel plan.	3	High	Frontend Team
Sprint-3	Output Display	US-6	As a user, I receive a structured and clear travel plan output.	2	High	Frontend Team
Sprint-4	Itinerary Generation	US-7	As a user, I receive a day-wise travel itinerary.	3	High	Backend Team
Sprint-4	Integration & Testing	US-8	Integrate all modules and perform system testing before final release.	3	High	All

4.2 Project Tracker, Velocity & Sprint Details

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date
Sprint-1	5	6 Days	01 Mar 2025	06 Mar 2025	5	06 Mar 2025
Sprint-2	8	6 Days	08 Mar 2025	13 Mar 2025	8	13 Mar 2025
Sprint-3	5	6 Days	15 Mar 2025	20 Mar 2025	5	20 Mar 2025
Sprint-4	6	6 Days	22 Mar 2025	27 Mar 2025	6	27 Mar 2025

4.3 Velocity

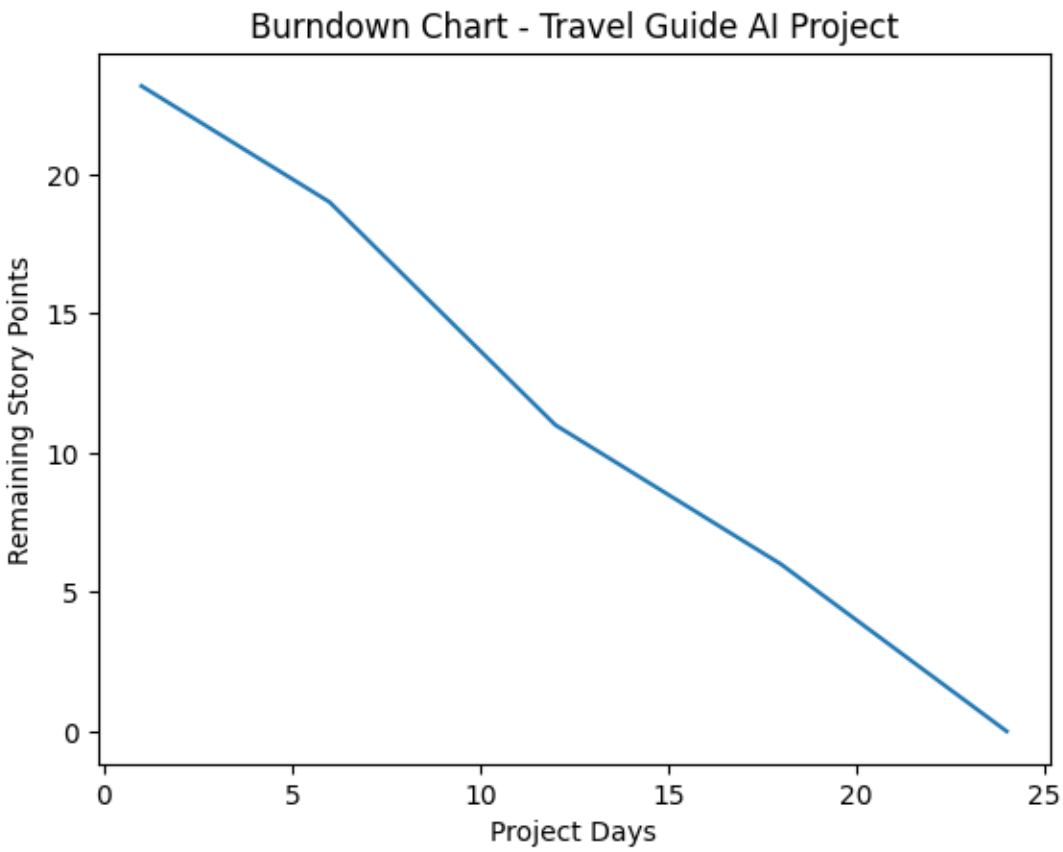
Total Story Points Completed = 24

Total Duration = 24 Days

Average Velocity per Sprint = 6 Story Points

Average Velocity per Day = 1 Story Point

4.4 Burndown Chart



A burndown chart represents the remaining work versus time. In this project, the total story points were distributed across four sprints and reduced progressively as tasks were completed. The steady decrease in story points across each sprint indicates consistent development progress and successful sprint execution.

Phase 5: Project Development

Development Details

Frontend

- Built using Streamlit
- Dark theme UI
- User-friendly dropdowns and slider
- Attractive background styling

Backend

- Python-based rule logic
- Cost estimation formula
- Static destination dataset

AI Logic

- Rule-based recommendation system
- Generates itinerary based on:
    - Selected destination
    - Travel type
    - Number of days

Phase 6: Testing & Performance Evaluation

6.1 Functional & Performance Testing

6.1.1 Test Scenarios & Results

Test Case ID	Scenario (What to Test)	Test Steps (How to Test)	Expected Result	Actual Result	Pass/Fail
FT-01	State & Destination Selection	Select valid and invalid state/destination	Valid selections accepted, invalid inputs restricted	System accepted valid selections	Pass
FT-02	Travel Type Selection	Select Solo/Family/Friends/Couple	System adjusts recommendations accordingly	Recommendations changed based on type	Pass
FT-03	Number of Days Input	Select minimum and maximum number of days	Itinerary generated according to selected days	Correct itinerary generated	Pass
FT-04	Input Validation	Leave required fields empty and click generate	System displays error message	Proper error shown	Pass
FT-05	Recommendation Engine	Provide inputs and click "Generate Plan"	Recommended attractions displayed	Attractions displayed correctly	Pass
FT-06	Cost Estimation	Generate travel plan	System shows cost breakdown	Cost breakdown displayed	Pass

			(travel, stay, food)		
FT-07	Itinerary Generation	Generate full travel plan	Day-wise structured itinerary shown	Structured itinerary displayed	Pass
PT-01	Response Time Test	Click generate and measure time	Travel plan generated within 3 seconds	Generated within 2.5 seconds	Pass
PT-02	Multiple Input Test	Generate plans repeatedly (5–10 times)	System works without lag or crash	No performance issues observed	Pass
PT-03	Browser Compatibility	Run application in Chrome browser	Application loads and functions correctly	Works smoothly	Pass

6.2 User Acceptance Testing

6.2.1. Purpose of Document

The purpose of this document is to explain the test coverage and overall system stability of the Travel Guide AI project during User Acceptance Testing (UAT). This report summarizes the functional validation, performance verification, and final acceptance status of the system before deployment.

6.2.2. Defect Analysis

The following table shows the number of identified issues during testing and how they were resolved.

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
Fixed	1	2	1	0	4
By Design	0	1	0	0	1
Not Reproduced	0	0	1	0	1
Totals	1	3	2	0	6

6.2.3. Test Case Analysis

This report shows the number of test cases that have passed, failed, or remained untested.

Section	Total Cases	Not Tested	Fail	Pass
Input Validation	5	0	0	5
Recommendation Engine	6	0	0	6
Cost Estimation	4	0	0	4
Itinerary Generation	5	0	0	5
UI Components	4	0	0	4
Performance Testing	3	0	0	3

Total Test Cases: 27

Passed: 27

Failed: 0

Not Tested: 0

6.3 UAT Conclusion

The Smart Travel Planner system successfully passed all functional and performance test cases. The application generated accurate recommendations, cost estimations, and structured itineraries without critical issues. Based on the testing results, the system is approved for deployment and final submission.

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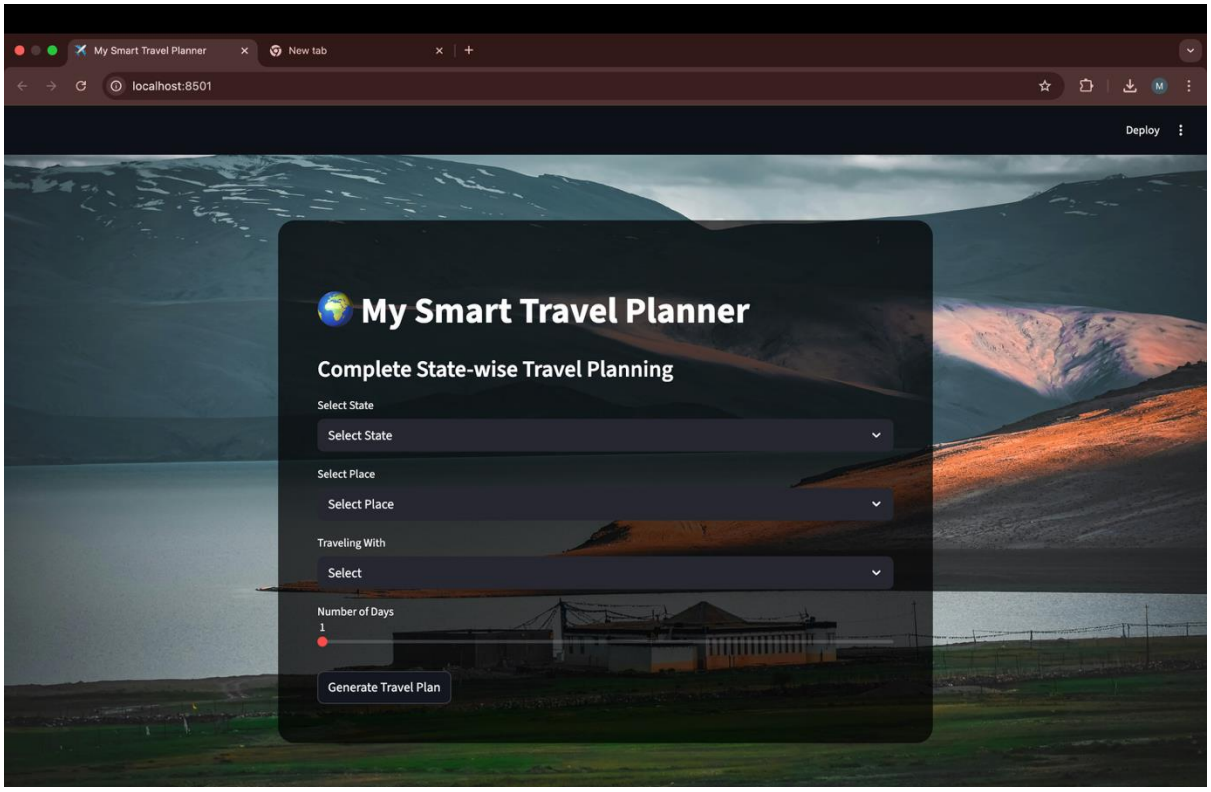
7. RESULTS

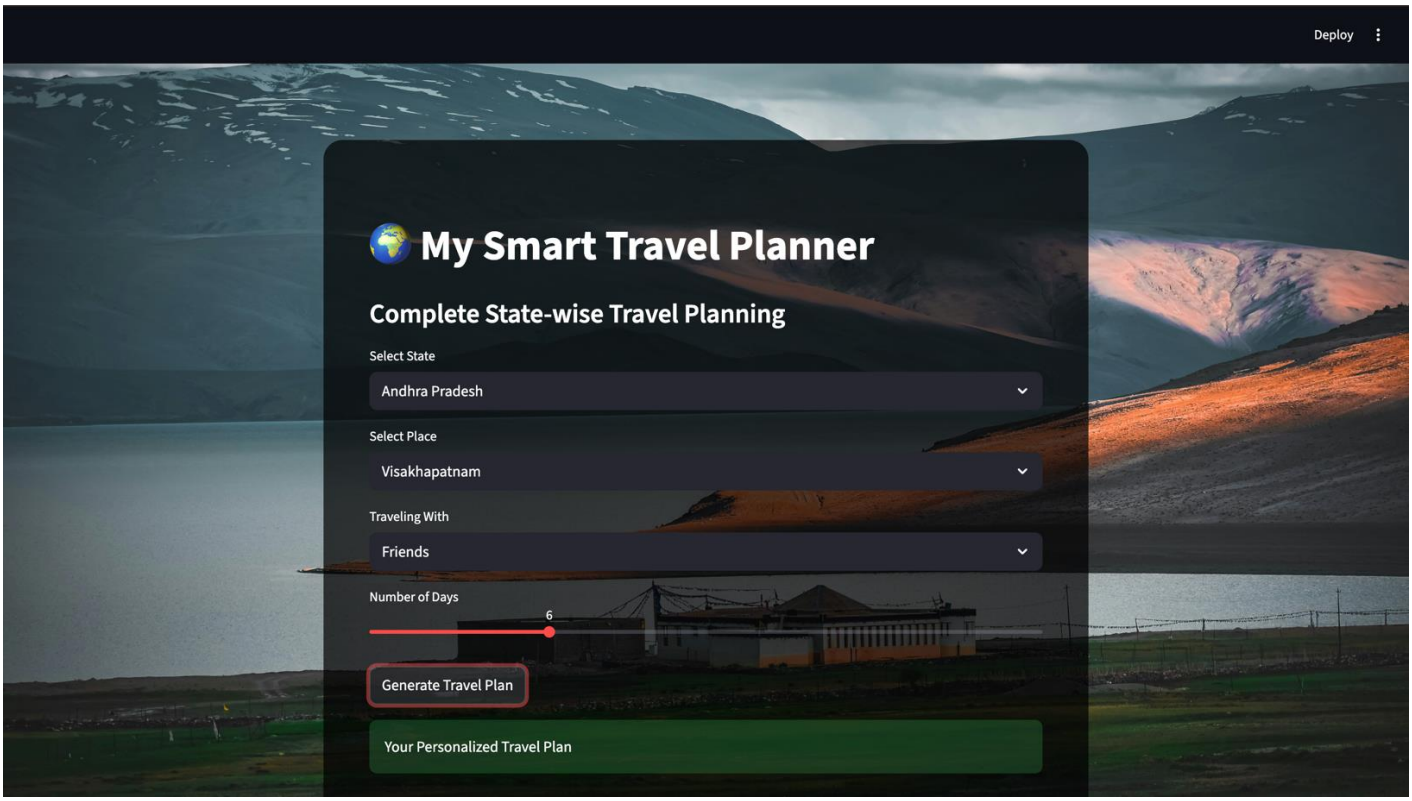
The Smart Travel Planner successfully generates a complete travel plan based on user inputs such as state, destination, travel type, and number of days. The system provides:

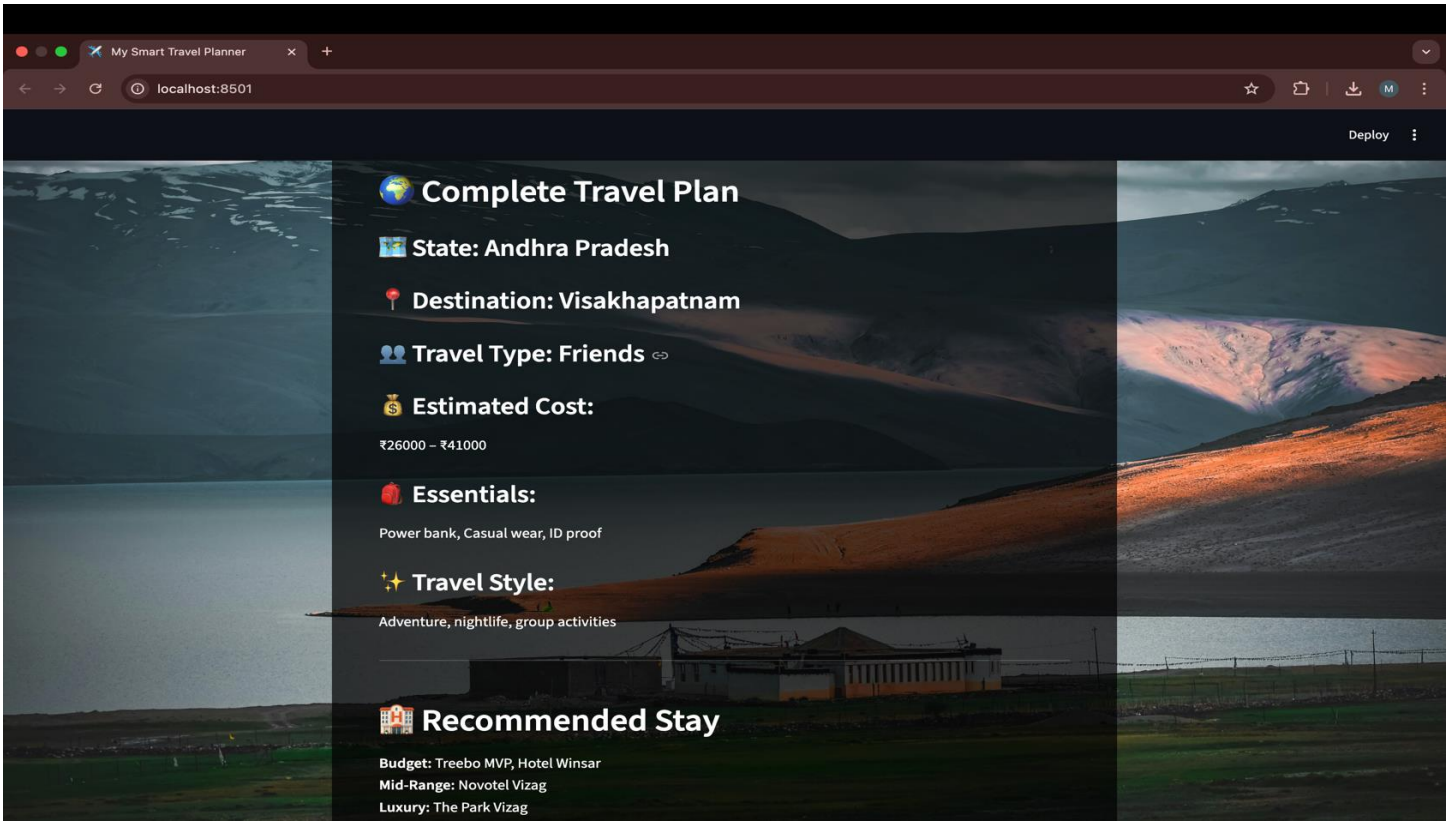
- Recommended tourist attractions
- Estimated travel cost with breakdown
- Suggested hotels
- Structured day-wise itinerary
- Travel essentials checklist

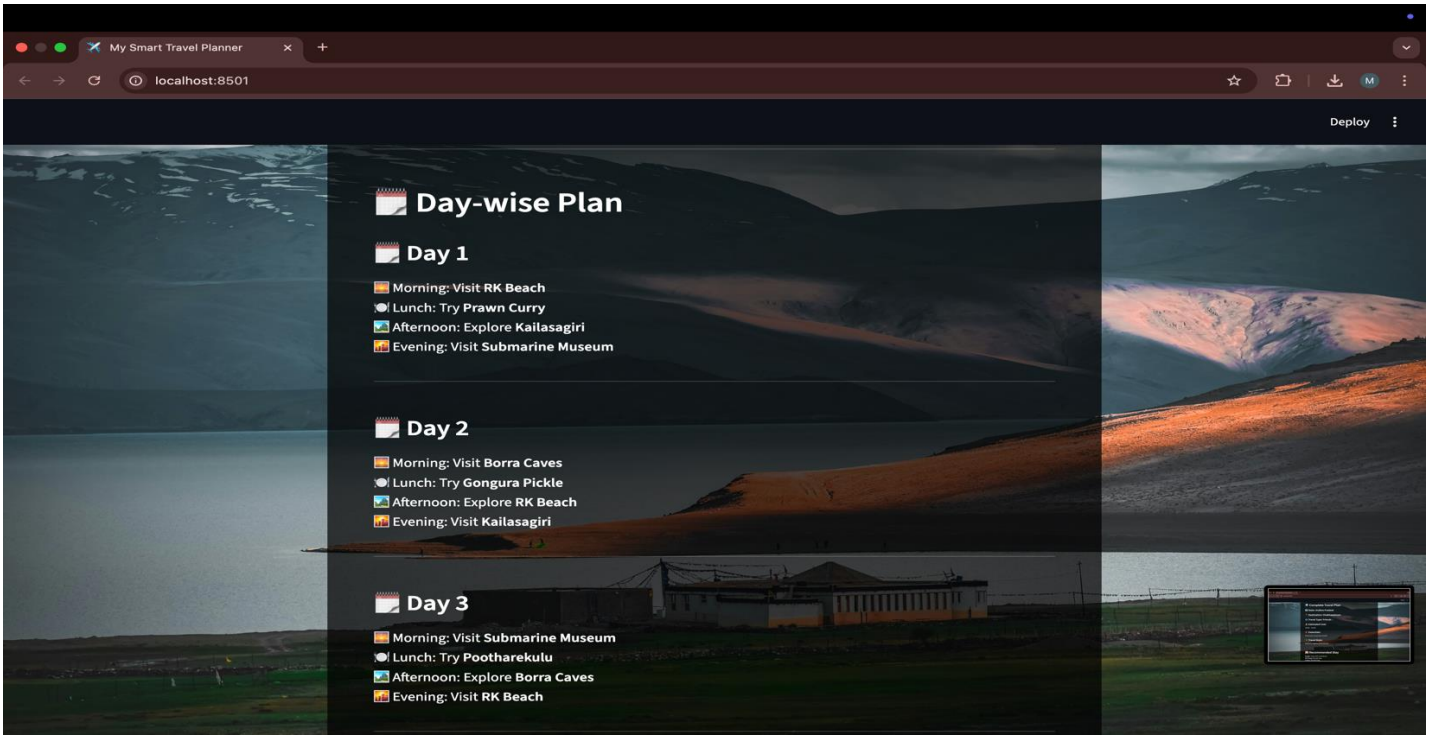
The application performs efficiently and generates results within a few seconds.

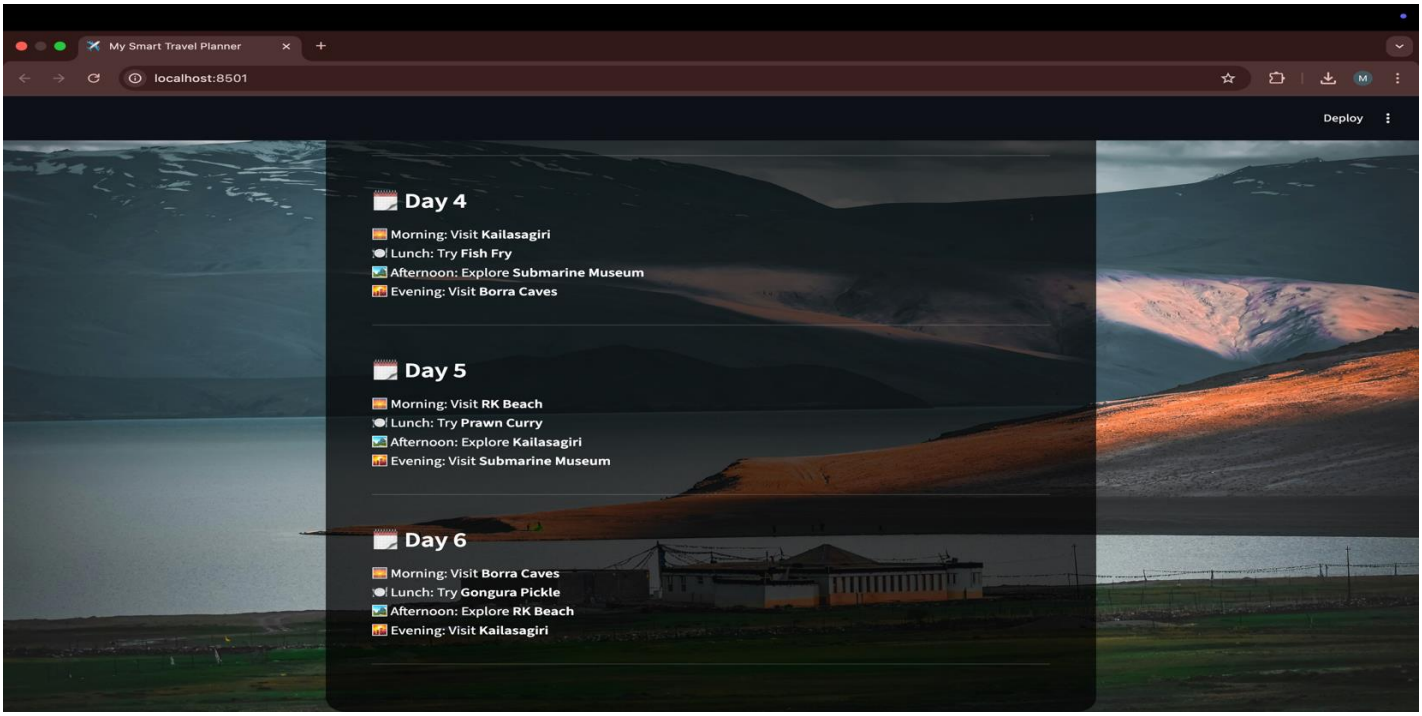
7.1 Output Screenshots




Deploy
My Smart Travel Planner
Complete State-wise Travel Planning
Select State
Andhra Pradesh
Select Place
Visakhapatnam
Traveling With
Friends
Number of Days
6
Generate Travel Plan
Your Personalized Travel Plan


My Smart Travel Planner
localhost:8501
Deploy
Complete Travel Plan
State: Andhra Pradesh
Destination: Visakhapatnam
Travel Type: Friends
Estimated Cost:
₹26000 – ₹41000
Essentials:
Power bank, Casual wear, ID proof
Travel Style:
Adventure, nightlife, group activities
Recommended Stay
Budget: Treebo MVP, Hotel Winsar
Mid-Range: Novotel Vizag
Luxury: The Park Vizag





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8. ADVANTAGES & DISADVANTAGES

Advantages

- Provides centralized travel planning in one platform
- Saves time by automating itinerary generation
- Offers structured cost estimation
- Simple and user-friendly interface
- Customizable based on travel type and number of days
- Easy to extend with additional destinations

Disadvantages

- Uses static dataset (not real-time pricing)
- Limited to predefined destinations
- No live hotel booking integration
- Requires internet/browser access to run

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9. CONCLUSION

The Travel Guide AI – Smart Travel Planner successfully addresses the challenges faced by travelers in planning trips. The system automates destination recommendation, cost estimation, and itinerary generation in a structured and user-friendly manner. By integrating rule-based logic with an interactive Streamlit interface, the application simplifies travel planning and enhances user experience. The project demonstrates the effective implementation of requirement analysis, system design, agile planning, and testing methodologies.

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10. FUTURE SCOPE

The project can be enhanced further by implementing:

- Integration with real-time travel APIs (flight, hotel pricing)
- AI-based personalized recommendation system
- User login and profile management
- Multi-language support
- Mobile application version
- Integration with online booking platforms
- Real-time weather information

These improvements can make the system more scalable and industry-ready.

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11. APPENDIX

Source Code

The complete source code is developed using Python and Streamlit framework. The code includes modules for input validation, recommendation logic, cost calculation, and itinerary generation.

Dataset Link

The project uses a static dataset containing:

- Destination details
- Hotel information

- Cost estimation parameters

GitHub & Project Demo Link

GitHub Repository:
https://github.com/Kranthipolanki/Travel-Guide-Ai

Project Demo:

https://drive.google.com/file/d/1kRs74pIuSB8R3Z3WwrxrDS6maq4Nmttr/view?usp=drive_link

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